$V_p = 120$ V RMS (170 V AMP) $\quad V_s = 26.25$ V RMS $\quad f = 100$ KHz

turns ratio $\quad \frac{V_p}{V_s} = \frac{120}{26.25} = 4.571$

Choose $\frac{N_p}{N_s} = 4.571$

Ferrite EFD 25 core $\rightarrow A_L = 2000$ nH/turns^2, $A_e = 58$ mm^2

EFD14 core $\rightarrow A_e = 173$ mm^2, $A_L = 3500$ nH/turn^2

$B_{pk} \leq 0.12$T (target flux density)

$$N_p \geq \frac{V_{p,AMP}}{4 f A_e B_{pk}} = \frac{170}{4(100k)(173mm^2)(0.12T)} \rightarrow N_p \geq 20.47$$

$\frac{N_p}{N_s} = \frac{V_p}{V_s} = \frac{120}{26.25} = \frac{32}{7}$, $N_p \geq 20.47$ so $\boxed{N_p = 32,\ N_s = 7}$

✓

$$B_{pk} = \frac{170}{4 f N_p A_e} = \frac{170}{4(100k)(32)(173mm^2)} = 0.076 < 0.12T$$

$$L_p = A_L N_p^2 = (3500n)(32^2) = \boxed{3.58 \text{ mH}}$$

$$I_p = \frac{V_{p,rms}}{2\pi f L_p} = \frac{120}{2\pi(100k)(3.58m)} = \boxed{53.3 \text{ mA}}$$

$$L_s = L_p \left(\frac{N_s}{N_p}\right)^2 = (3.58m)\left(\frac{7}{32}\right)^2 = \boxed{171 \text{ pH}}$$